

Due Friday, May 15th 2026 (11:59PM)**Problem 1. (20 pts)** Consider the following relational database:*employee* (employee-name, street, city)*works* (employee-name, company-name, salary)*company* (company-name, city)*manages* (employee-name, manager-name)

Write the following queries in SQL:

- Find the names of employees who work in a city different from the city where their company is located.
- Find the names of employees who earn less than the average salary of their company.

Problem 2. (60pts) Write the following queries in SQL on the database schema shown below.

- For departments with an average salary greater than \$50,000, list the department name and the average salary of employees in each department.
- Find the first and last names of employees who do not work on any project.
- Retrieve the last names of employees who share a birthday with at least one of their dependents.
- Find the first and last names of employees who work on every project in the "Houston" location.
- For each department, list the department name and the number of employees who work in that department.
- List the last names of all employees who have no dependents.

EMPLOYEE

FNAME	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNUM
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DEPARTMENT

DNAME	<u>DNUMBER</u>	MGRSSN	MGRSTARTDATE
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DEPT_LOCATIONS

<u>DNUMBER</u>	<u>DLOCATION</u>
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WORKS_ON

<u>ESSN</u>	<u>PNUM</u>	HOURS
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PROJECT

PNAME	<u>PNUMBER</u>	PLOCATION	DNUM
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DEPENDENT

<u>ESSN</u>	<u>DEPENDENT_NAME</u>	SEX	BDATE	RELATIONSHIP
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Problem 3. (20pts) Consider the following relational schema for a university database:

STUDENT (Name, StudentNumber, Class, Major)
COURSE (CourseName, CourseNumber, CreditHours, Department)
PREREQUISITE (CourseNumber, PrerequisiteNumber)
SECTION (SectionIdentifier, CourseNumber, Semester, Year, Instructor)
GRADE_REPORT (StudentNumber, SectionIdentifier, Grade)

Write the queries below in SQL:

1. Retrieve the names and majors of all 'A' students (i.e., students who have a grade of 'A' in all their courses).
2. Retrieve the names and majors of all students who do not have a grade of 'A' in any of their courses.